



MARMARA
ÜNİVERSİTESİ

BIOMETRIC IDENTITY VERIFICATION, AS A CLOUD SERVICE



ACTIVE-LIVENESS ANTI-SPOOFING · TEN COMPOSABLE AUTH FACTORS · MULTI-TENANT SAAS

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Advisor: Assoc. Prof. Dr. Mustafa Ağaoğlu · CSE4197 / CSE4198 · Spring 2025–2026

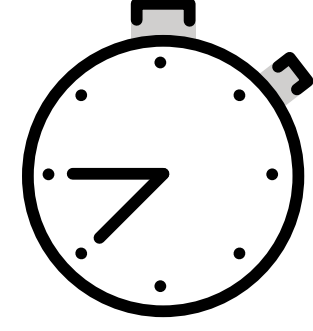
Marmara University · Faculty of Engineering · Department of Computer Engineering



0 %

replay attacks blocked

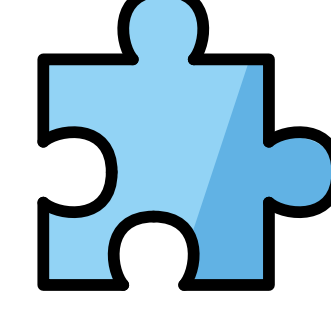
0 / 120 recorded-video attempts



< 1 s

end-to-end face verification

P95 = 950 ms on a commodity CPU

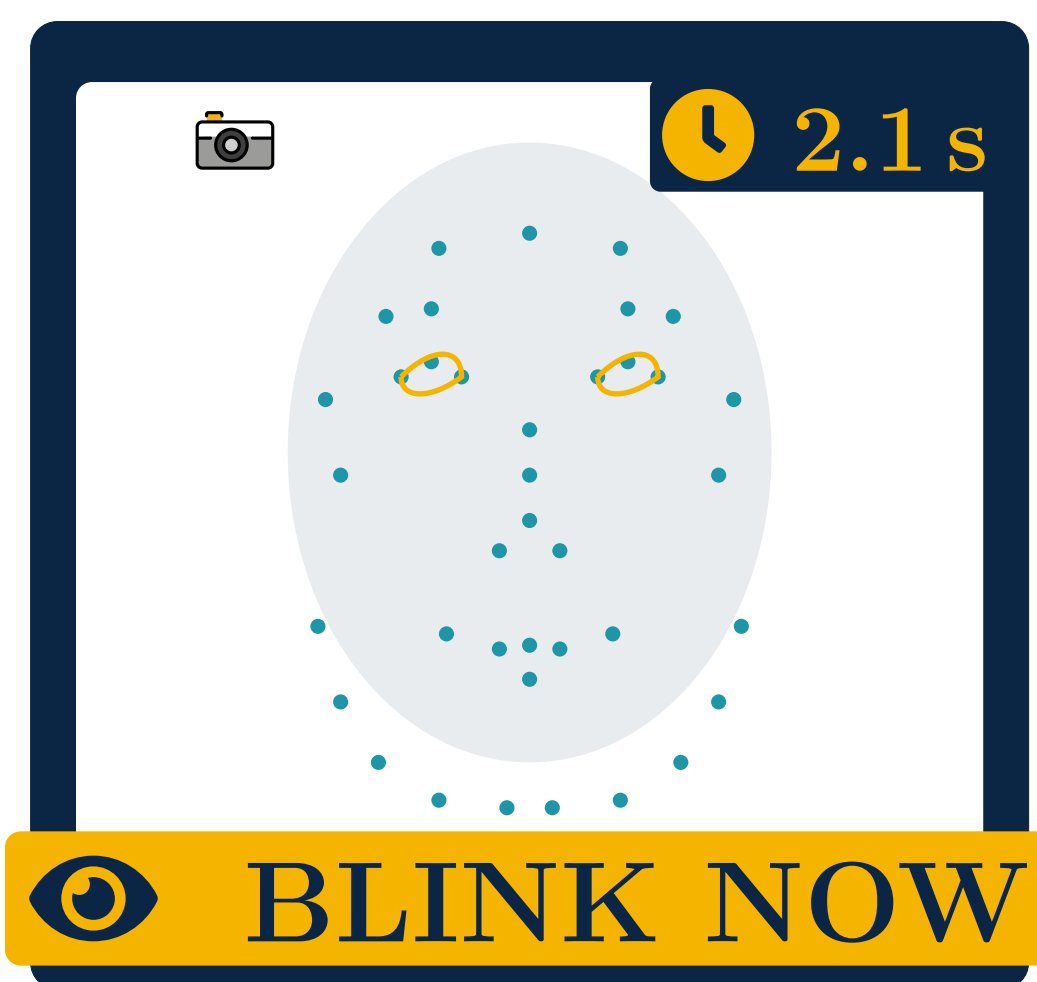


10

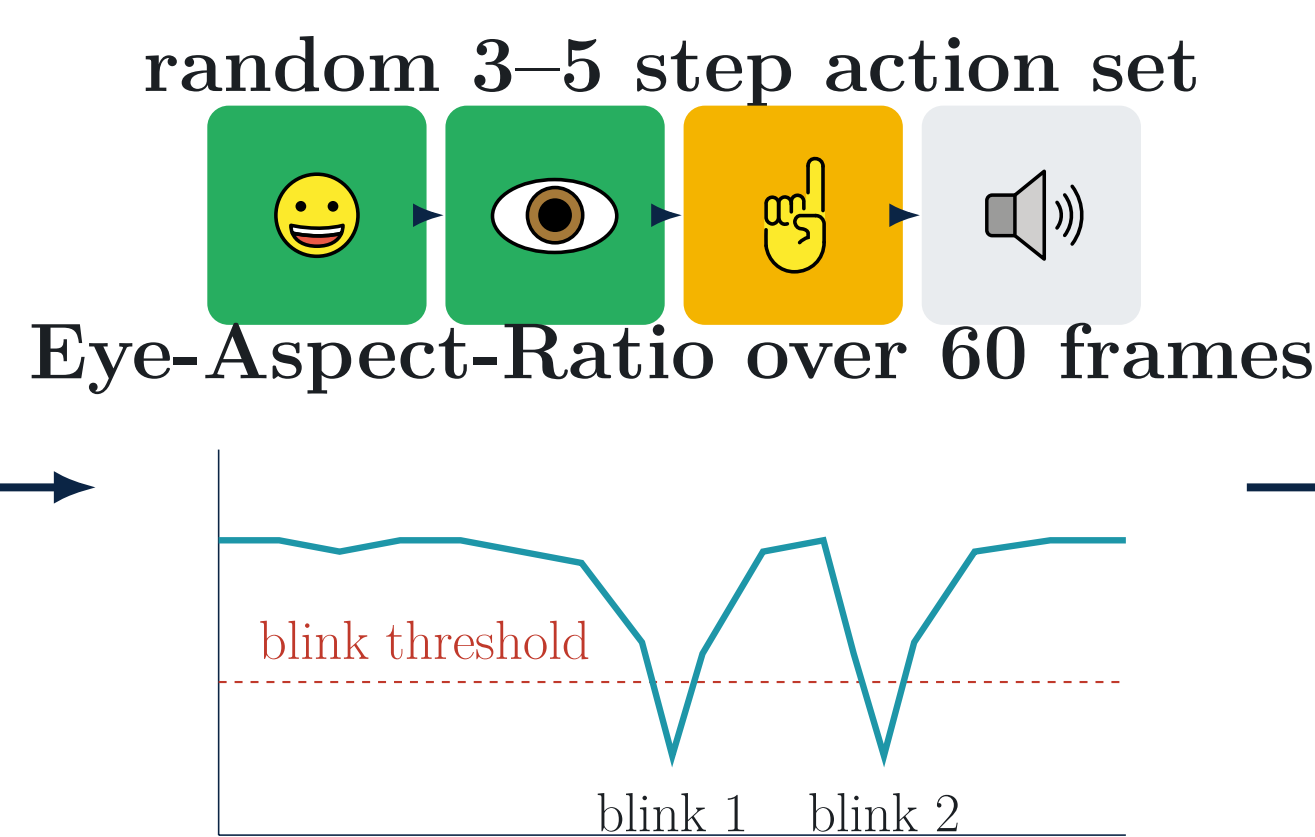
composable auth factors

any subset → a tenant MFA flow

How it works — capture, challenge, verdict



1 Capture face mesh



2 Pass the random challenge

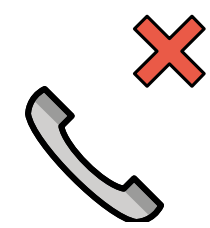


3 Encrypt & return verdict

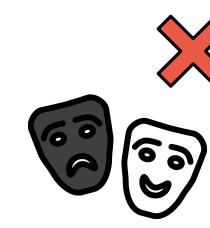
The problem



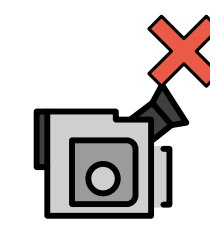
passwords



SMS OTP



deepfake



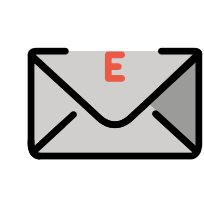
replay clip

One passive signal — a selfie or a clip — is all a prepared attacker needs.

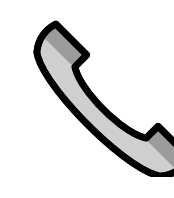
Ten composable auth factors



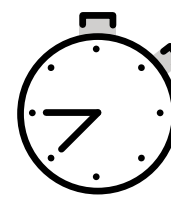
Password



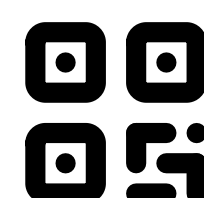
Email OTP



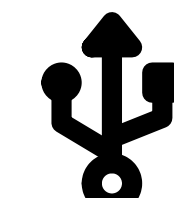
SMS OTP



TOTP



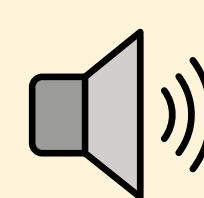
QR code



WebAuthn



Face



Voice



Fingerprint



NFC document



Gold tiles =
active liveness

Where the second goes

950 ms

end-to-end P95 face verification, single CPU server

120

180

240

210

200

detection

face mesh

embedding

anti-spoof

vector search

Proven on the live deployment



0 / 120
replay
false-accepts

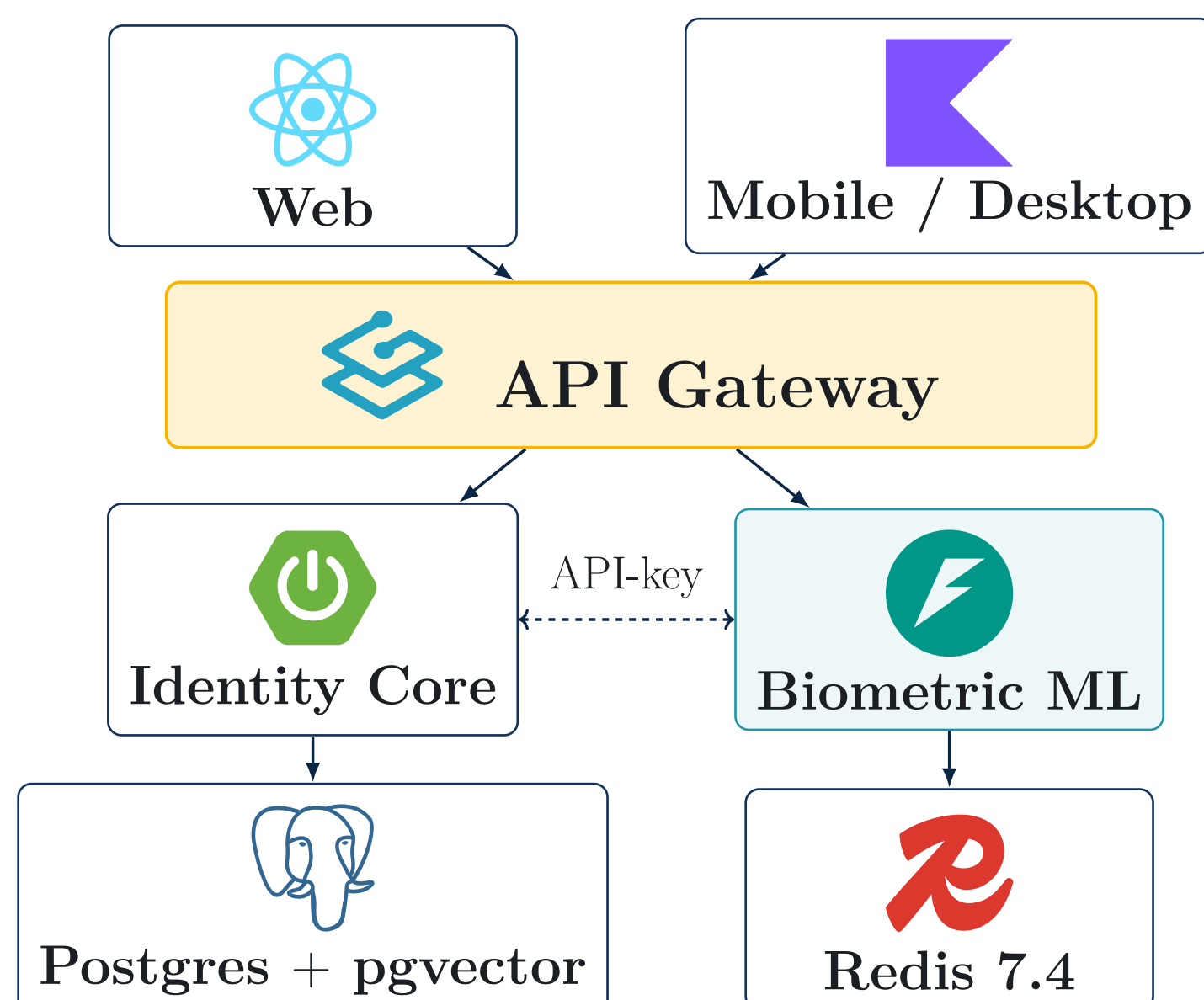


0.97
passive-spoof
AUC



1820+
tests gate
every PR

Architecture



Biometric ML has **no public route** — private Docker network only.

Engineering at a glance



60

Flyway migrations



1820+

automated tests



5

Git submodules

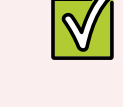
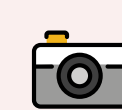


8

architecture ADRs

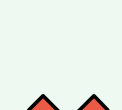
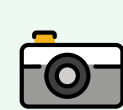
Passive vs. active liveness

Passive only



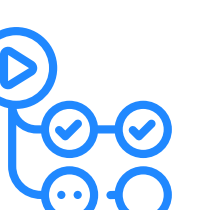
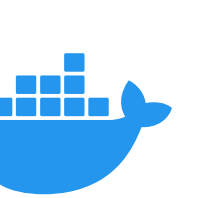
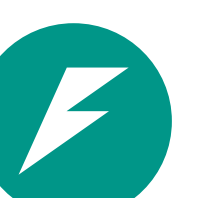
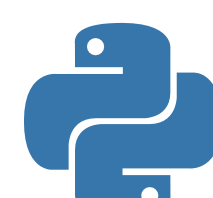
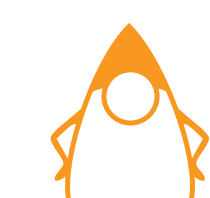
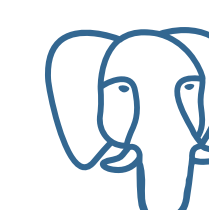
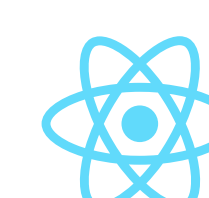
photo, clip & deepfake all pass

Active challenge

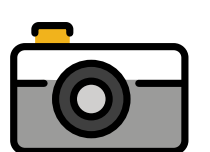


random sequence blocks all three

Built with



See it live



Scan to try a
verification

verify.fivucsas.com